

[Menu](#)

Search Interface

▼ Relevant source files

`h4styles.css``hclient/core/hapi.js``hclient/core/recordset.js``hclient/core/utils.js``hclient/core/utils_dbs.js``hclient/core/utils_msg.js``hclient/core/utils_ui.js``hclient/widgets/editing/editing2.js``hclient/widgets/editing/editing_exts.js``hclient/widgets/editing/editing_input.js``hclient/widgets/entity/manageDefDetailTypes.js``hclient/widgets/entity/manageDefRecStructure.js``hclient/widgets/entity/manageDefRecTypes.js``hclient/widgets/entity/manageDefTerms.js``hclient/widgets/entity/manageEntity.js``hclient/widgets/entity/manageRecords.js``hclient/widgets/search/search.js``hclient/widgets/search/searchBuilder.html``hclient/widgets/search/searchBuilder.js``hclient/widgets/search/searchBuilderInterface.js``hclient/widgets/search/searchBuilderSort.js``hclient/widgets/search/search_faceted.js`

```
hclient/widgets/search/search_faceted_wiz.html
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hclient/widgets/search/search_faceted_wiz.js
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hclient/widgets/search/svs_edit.html
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hclient/widgets/search/svs_edit.js
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hclient/widgets/search/svs_list.js
```

```
hclient/widgets/viewers/resultList.js
```

This document describes the search capabilities in the Heurist system, including basic search, faceted search, and search results display. The search interface allows users to find records based on various criteria, filter results, save search configurations, and interact with search results.

For information about record editing based on search results, see [Record Editing](#).

Overview

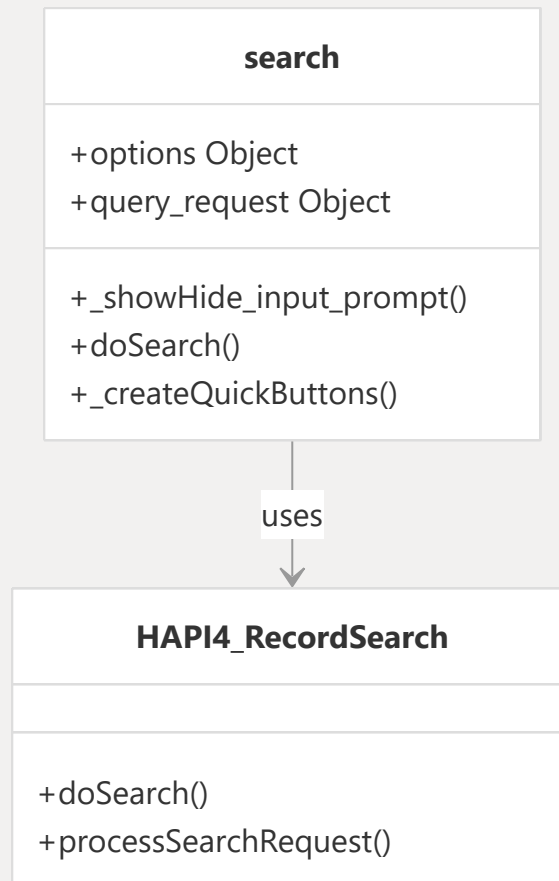
The Heurist search interface comprises several interconnected components that work together to provide comprehensive search functionality:

Sources:

- `hclient/widgets/search/search.js` 20-60
- `hclient/widgets/search/search_faceted.js` 33-134
- `hclient/core/hapi.js` 447-496
- `hclient/widgets/viewers/resultList.js` 23-70

Basic Search

The basic search component provides a simple text input that allows users to search across records. It supports both simple text queries and more complex structured queries.



The search widget (`search.js`) handles:

- Input validation
- Domain selection (all records, bookmarks, etc.)
- Query preparation
- Triggering search events

Key options include:

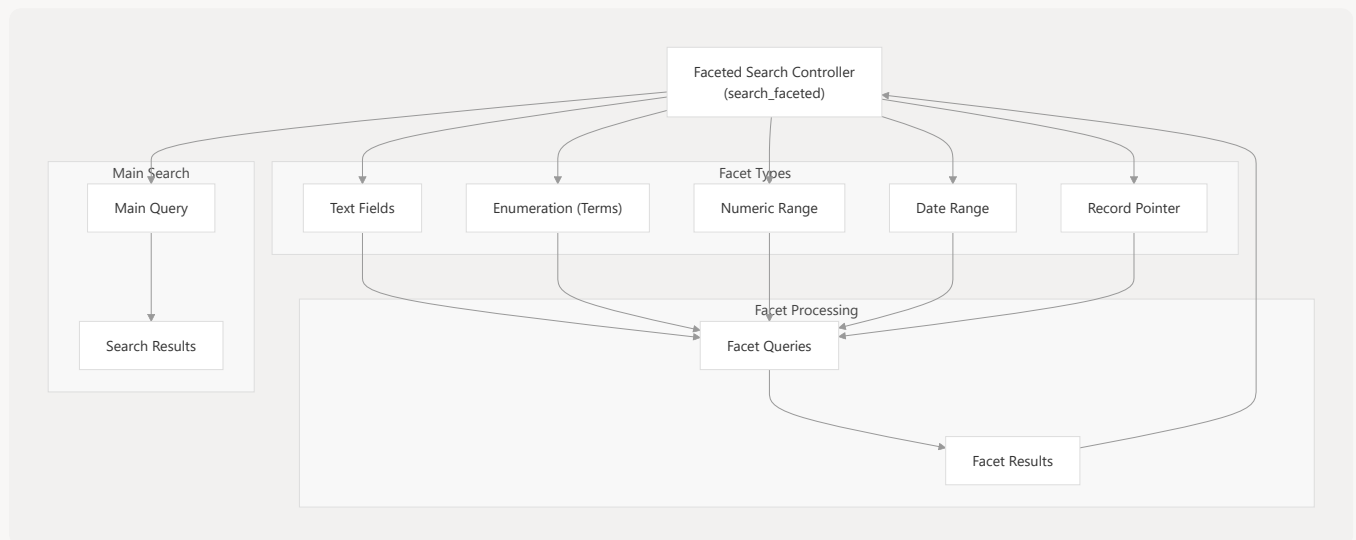
- `search_domain` : Defines the scope of the search ('a' for all, 'b' for bookmarks, etc.)
- `search_realm` : Used to group related search components
- `search_button_label` : Customizes the search button text
- `search_input_label` : Customizes the input field label

Sources:

- `hclient/widgets/search/search.js` 25-66
- `hclient/widgets/search/search.js` 148-214
- `hclient/core/hapi.js` 453-496

Faceted Search

Faceted search provides a more structured approach to searching by presenting predefined categories (facets) that users can filter by. This allows users to narrow down search results incrementally.



The faceted search system:

1. Defines facets based on record structure
2. Presents facet values for selection
3. Builds complex queries based on facet selections
4. Updates available facet values based on current selection

Key concepts:

- **Facet Types:** Different field types (text, enum, numeric, date, record pointers)
- **Facet Rendering Modes:** Different display methods (dropdown, list, range selectors)
- **Linked Facets:** Facets that follow relationship paths to related records
- **Facet Recalculation:** Dynamically updating available facet values

Sources:

- `hclient/widgets/search/search_faceted.js` 136-207
- `hclient/widgets/search/search_faceted.js` 232-343
- `hclient/widgets/search/search_faceted_wiz.js` 81-135

Facet Query Structure

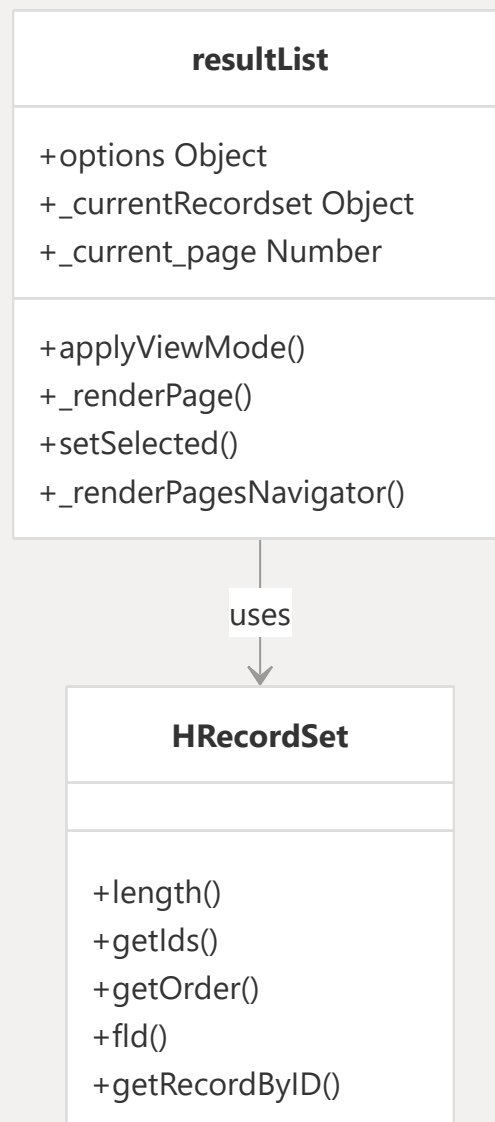
Faceted search uses a sophisticated query structure to handle both direct field queries and linked record queries:

Sources:

- `hclient/widgets/search/search_faceted.js` 34-94
- `hclient/widgets/search/search_faceted_wiz.js` 28-77

Search Results

The `resultList` widget handles the display of search results, providing various view modes and interactions with the result items.



Key features of the result list:

- Multiple view modes (list, icons, thumbnails, etc.)
- Record selection (single/multi)
- Pagination
- Sorting and filtering
- Record preview
- Action buttons for record operations

The result list listens to search events and updates its display accordingly:

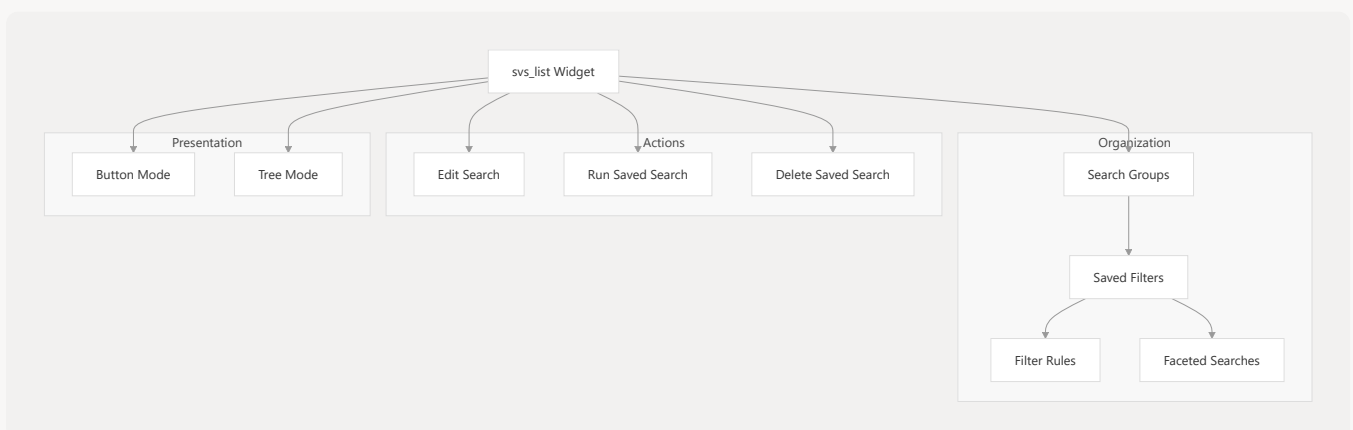
- **ON_REC_SEARCHSTART** : Prepares for new search results
- **ON_REC_SEARCH_FINISH** : Processes and displays search results
- **ON_REC_SELECT** : Handles record selection events

Sources:

- `hclient/widgets/viewers/resultList.js` 23-147
- `hclient/widgets/viewers/resultList.js` 251-474
- `hclient/core/recordset.js` 1-17

Saved Searches

The **svs_list** widget manages saved searches, allowing users to save, organize, and reuse search configurations.



The saved search system provides:

- Organization of searches into groups
- Different types of saved searches (filters, rules, faceted)
- Different display modes (button mode, tree mode)
- Search editing and management

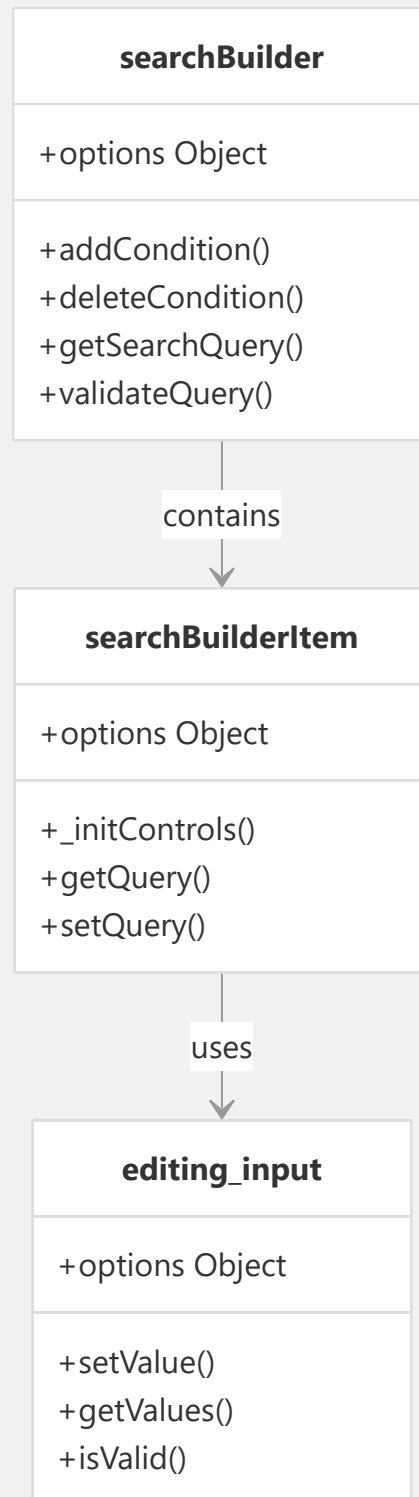
- Support for favorites

Sources:

- `hclient/widgets/search/svs_list.js` 22-60
- `hclient/widgets/search/svs_list.js` 82-115
- `hclient/widgets/search/svs_edit.js` 1-10

Advanced Search Builder

The advanced search builder provides a visual interface for constructing complex queries with multiple conditions, operators, and groupings.



The search builder supports:

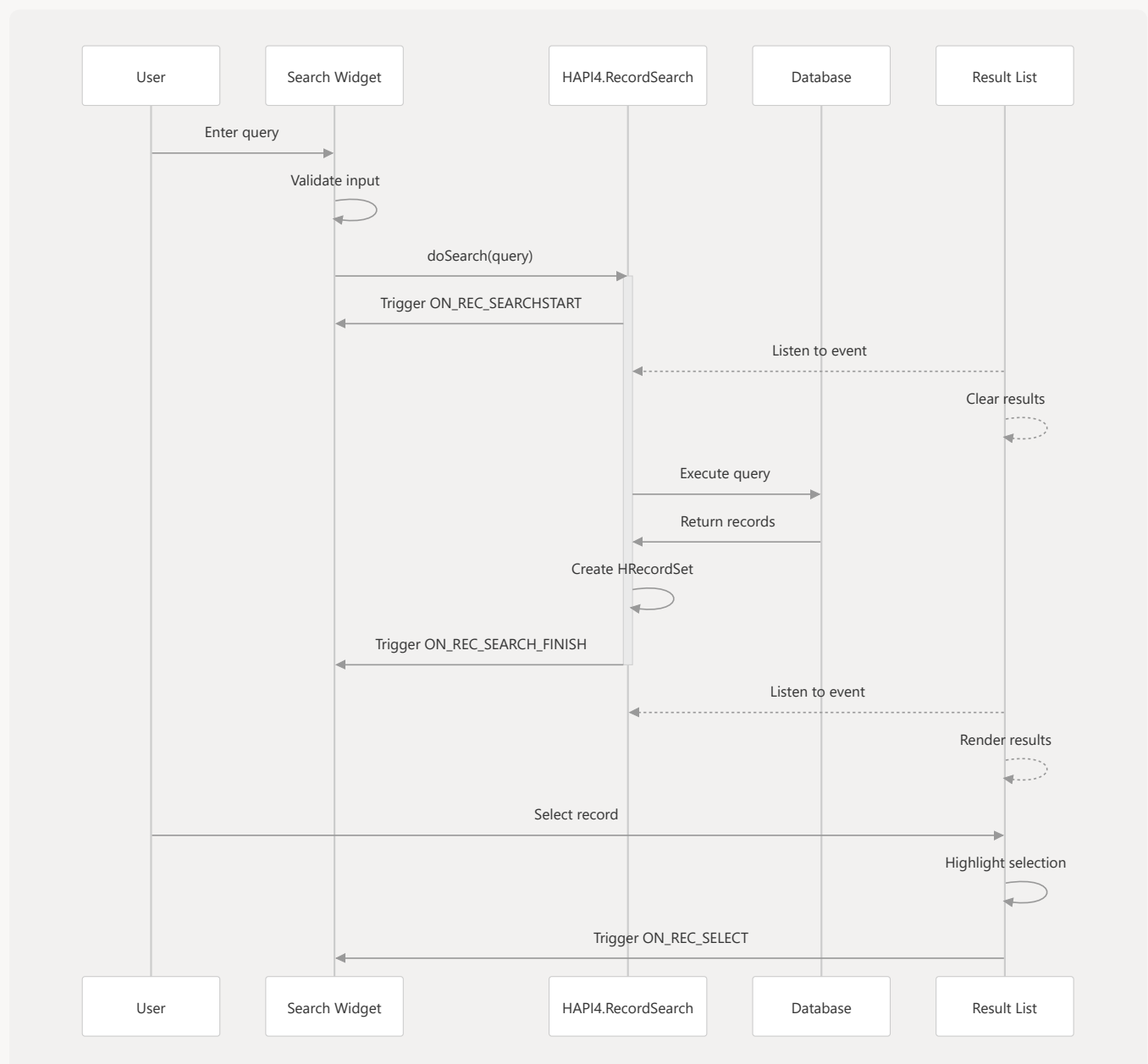
- Multiple conditions with different field types
- Logical operators (AND, OR, NOT)
- Grouping of conditions
- Field-specific input controls
- Query validation
- Saving and loading queries

Sources:

- `hclient/widgets/search/searchBuilder.js` 11-14
- `hclient/widgets/search/searchBuilderItem.js` 1-10
- `hclient/widgets/editing/editing_input.js` 23-26

Search Flow and Events

The search process uses a system of events to coordinate between components. This allows different parts of the interface to react to search actions.



Key events:

- **ON_REC_SEARCHSTART** : Indicates a search is beginning
- **ON_REC_SEARCH_FINISH** : Indicates search results are available

- `ON_REC_SELECT` : Indicates record selection has changed
- `ON_REC_UPDATE` : Indicates records have been modified

Search events include information about:

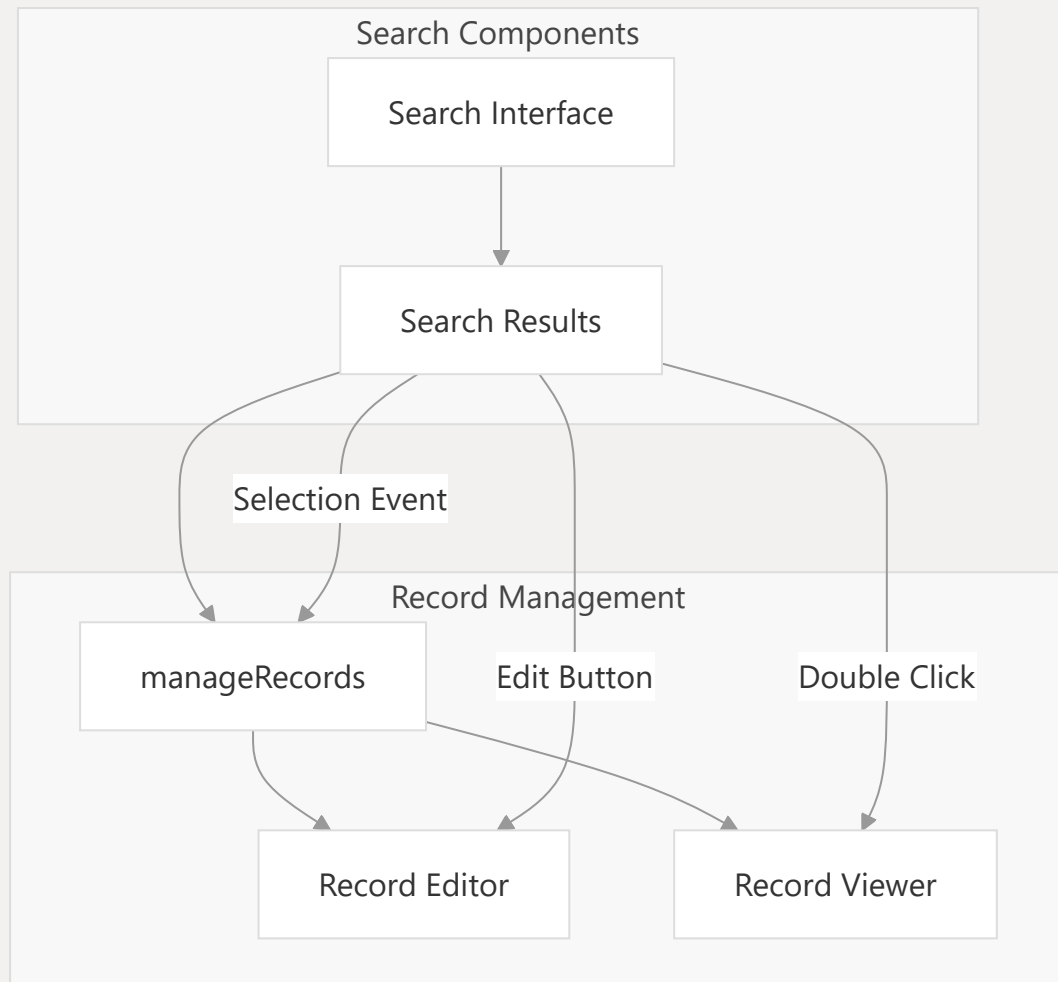
- Search query
- Result set
- Selected records
- Source component
- Search realm (for grouping related components)

Sources:

- `hclient/widgets/search/search.js` 284-304
- `hclient/widgets/viewers/resultList.js` 251-376
- `hclient/core/hapi.js` 376-422

Integration with Record Management

The search interface integrates closely with the record management system, providing ways to view, edit, and manage records found in search results.



Integration points:

- Result list provides actions to open records in editors or viewers
- Record editors can trigger searches
- Search results can be passed to record management components
- Events coordinate between search and record management

Sources:

- `hclient/widgets/entity/manageRecords.js` 22-58
- `hclient/widgets/viewers/resultList.js` 94-116

Search Configuration and Customization

The search interface offers numerous configuration options to customize behavior and appearance.

Basic Search Configuration

```
search: {
  search_domain: 'a',           // Search domain (a=all, b=bookmarks, etc.)
  search_domain_set: 'a,b,r,s', // Available domains
  btn_visible_newrecord: true,   // Show add record button
  btn_entity_filter: true,      // Show entity filter buttons
  search_button_label: 'Search', // Custom button label
  search_input_label: 'Filter',  // Custom input label
  search_realm: 'main'          // Group related search components
}
```

Faceted Search Configuration

```
search_faceted: {
  params: {
    viewport: 5,           // Number of items shown before "more" button
    accordion_view: true,  // Use accordion for facets
    ui_additional_filter: true // Allow additional text filter
  },
  svcs_ID: 123,           // ID of saved faceted search
  hide_no_value_facets: true, // Hide facets with no available values
  search_realm: 'main'     // Group related search components
}
```

Result List Configuration

```
resultList: {
  view_mode: 'list',           // Display mode (list, icons, thumbs, etc.)
  select_mode: 'select_multi', // Selection mode
  multiselect: true,           // Allow multiple selection
  show_toolbar: true,          // Show toolbar
  show_menu: true,             // Show menu
  pagesize: 50,                // Results per page
  recordview_onselect: 'popup', // Preview behavior on select
  search_realm: 'main'         // Group related search components
}
```

Sources:

- `hclient/widgets/search/search.js` 27-50
- `hclient/widgets/search/search_faceted.js` 150-167
- `hclient/widgets/viewers/resultList.js` 26-146

Technical Reference

Key Search Classes and Components

Component	File	Purpose
search	hclient/widgets/search/search.js	Main search input widget
search_faceted	hclient/widgets/search/search_faceted.js	Faceted search interface
search_faceted_wiz	hclient/widgets/search/search_faceted_wiz.js	Wizard for creating faceted searches
searchBuilder	hclient/widgets/search/searchBuilder.js	Advanced search builder
resultList	hclient/widgets/viewers/resultList.js	Search results display
svs_list	hclient/widgets/search/svs_list.js	Saved search management
HAPI4.RecordSearch	hclient/core/hapi.js	Core search processing
HRecordSet	hclient/core/recordset.js	Search results container

Search Events

Event	Description
ON_REC_SEARCHSTART	Triggered when a search begins
ON_REC_SEARCH_FINISH	Triggered when search results are available
ON_REC_SELECT	Triggered when record selection changes
ON_REC_STATUS	Triggered when record status changes
ON_REC_UPDATE	Triggered when records are updated
ON_REC_COLLECT	Triggered when records are collected

Key Search Methods

Method	Component	Purpose
doSearch	search	Initiates a search with the current query
_initFacetQueries	search_faceted	Creates facet search queries
_recalculateFacets	search_faceted	Updates facet values based on current selection
doSaveSearch	svs_edit	Saves the current search configuration
_renderRecordsIncrementally	resultList	Renders search results in the result list
doSearch	HAPI4.RecordSearch	Core method to execute search queries

Sources:

- `hclient/widgets/search/search.js` 284-304
- `hclient/widgets/search/search_faceted.js` 232-343
- `hclient/core/hapi.js` 453-496
- `hclient/widgets/viewers/resultList.js` 429-446

Search Query Syntax

Heurist supports several query formats: